Notes: Hong & Kacperczyk (QJE 2010)

Competition and Bias

Contents

1 Big picture: what the paper is doing

1.1 Research question

The paper asks a causal question about *information supply*:

> Does more competition among information intermediaries reduce reporting bias?

The setting is U.S. **sell-side analysts' annual earnings forecasts**, which are known to be optimistic on average due to conflicts of interest (e.g., investment-banking ties, management access, career concerns).

1.2 Key empirical obstacle

Analyst coverage (a proxy for competition) is *endogenous*. Stocks that attract more analysts differ systematically from those that do not, and analyst coverage is persistent and responds to firm performance. Therefore, OLS regressions of forecast bias on coverage have ambiguous interpretation.

1.3 Identification idea (natural experiment)

The paper uses **brokerage-house mergers** as a natural experiment for competition. When two brokerage houses merge, *redundant analysts are often let go* (and coverage contracts). The core treatment group is the set of stocks *covered by both merging houses pre-merger* (overlapping coverage). These stocks face a plausibly exogenous decline in the number of competing analysts after the merger.

1.4 Main findings (economic content)

- Mergers lead to a **drop in coverage** for the treatment stocks of about **one analyst** (DID).
- The same treatment stocks experience an **increase in optimism bias** the year after the merger (DID), consistent with competition disciplining bias.
- The implied causal effect is **larger than OLS** estimates, consistent with OLS being attenuated/bias-contaminated by endogenous coverage.
- Effects are **much stronger for low-coverage stocks** (where competition is scarce to begin with).
- Auxiliary evidence: distributional shifts in bias (kernel densities) and cross-sectional incentive patterns in analysts' career outcomes.

2 Data, sample construction, and variable definitions

2.1 Data sources and sample

- Analyst forecasts: **IBES**, 1980–2005, focusing on **annual** earnings forecasts.

- Firm fundamentals: **COMPUSTAT**.
- Prices/returns: **CRSP**.

For each year t, the paper takes the *most recent* annual earnings forecast issued by each analyst for a stock, yielding one forecast per analyst–stock–year.

2.2 Forecast bias and related measures

Fix stock i and year t. Let analyst j's forecasted EPS be $\widehat{EPS}_{j,i,t}$ and the realized EPS be $EPS_{i,t}$ (from COMPUSTAT). Let $P_{i,t-1}$ be the prior-year stock price.

Analyst-level bias.

$$Bias_{j,i,t} \equiv \frac{\widehat{EPS}_{j,i,t} - EPS_{i,t}}{P_{i,t-1}}. \tag{1}$$

This is a *signed* forecast error scaled by price, so positive values mean optimistic forecasts.

Stock-level consensus bias. Let $\mathcal{J}_{i,t}$ denote the set of analysts covering i in year t. The paper uses both mean and median consensus bias:

$$BIAS_{i,t}^{mean} \equiv \frac{1}{|\mathcal{J}_{i,t}|} \sum_{j \in \mathcal{J}_{i,t}} Bias_{j,i,t}, \qquad BIAS_{i,t}^{med} \equiv \text{median}_{j \in \mathcal{J}_{i,t}} \{Bias_{j,i,t}\}. \tag{2}$$

Forecast error (absolute error).

$$FERROR_{j,i,t} \equiv \left| \frac{\widehat{EPS}_{j,i,t} - EPS_{i,t}}{P_{i,t-1}} \right|. \tag{3}$$

The paper also forms mean/median stock-level versions analogous to $BIAS$.

Dispersion.

$$FDISP_{i,t} \equiv \text{sd}_{j \in \mathcal{J}_{i,t}} \left(\widehat{EPS}_{j,i,t} \right), \tag{4}$$

the cross-analyst standard deviation of forecasts for the same stock-year.

2.3 Competition proxy: analyst coverage

$$COVERAGE_{i,t} \equiv |\mathcal{J}_{i,t}|, \tag{5}$$

the number of analysts covering firm i at the end of year t.

2.4 Controls used in the paper

Controls include (all as of $t-1$ when used as regressors):

- $\ln(SIZE)$: log market capitalization,
- $\ln(BM)$: log book-to-market,

- $RETANN$: annual return,
- σ: volatility of daily returns,
- $VOLROE$: volatility of return on equity (AR(1) residual variance),
- $PROFIT$: profitability (operating income / assets),
- $SP500$: indicator for S&P 500 membership,
- REC: recency of the forecast (distance from the earnings report date; used in some specifications).

2.5 Sample filters

Key filters used throughout:

- exclude stock-years with price below $5;
- exclude extreme stock-level bias observations (outer tails);
- exclude analyst forecasts with an absolute forecast–actual difference exceeding $10 (likely coding errors);
- OLS sample is restricted to large firms (top size quartile) to align with the natural-experiment sample.

3 Model 1: Baseline OLS relation between bias and coverage

3.1 Stock-year regression specification

A baseline stock-year regression consistent with Table II is:

$$BIAS_{i,t} = \alpha + \beta\, COVERAGE_{i,t-1} + \boldsymbol{\gamma}'\boldsymbol{X}_{i,t-1} + \lambda_t + \lambda_{\text{ind}(i)} + \varepsilon_{i,t}, \tag{6}$$

where $\boldsymbol{X}_{i,t-1}$ collects the controls, λ_t are year fixed effects, and $\lambda_{\text{ind}(i)}$ are industry fixed effects. Some specifications also include brokerage-house fixed effects and firm fixed effects.

3.2 Why OLS is not causal here

Coverage is chosen endogenously by analysts/brokerages and responds to (unobserved) expected earnings news, visibility, liquidity, and firm performance. Therefore, $\text{Cov}(COVERAGE_{i,t-1}, \varepsilon_{i,t}) \neq 0$ is plausible, making β difficult to interpret causally.

Direction of bias is ambiguous.

- If "exciting" stocks attract more analysts and are also associated with more optimistic forecasts, OLS can be biased *downward* (attenuated).
- If low-coverage stocks are "under-the-radar" and analysts initiate coverage only when very optimistic, OLS can be biased *upward*.

This motivates the natural experiment.

4 Natural experiment: brokerage-house mergers

4.1 Treatment vs control and event windows

For each brokerage-house merger m, define:

- **Treatment stocks:** covered by *both* merging houses in the year before the merger.
- **Control stocks:** not covered by both houses (various definitions; the paper uses matched controls).
- **Pre-window:** one-year period prior to the merger.
- **Post-window:** one-year period after the merger.

4.2 Model 2: Difference-in-differences (DID)

Let C be any characteristic (coverage, mean bias, median bias, etc.). Let $C_{T,1}$ and $C_{T,2}$ be the pre and post averages in the treatment group, and similarly $C_{C,1}$ and $C_{C,2}$ for controls. The DID estimator is:

$$DID \equiv (C_{T,2} - C_{T,1}) - (C_{C,2} - C_{C,1}). \tag{7}$$

4.2.1 Unbiasedness under parallel trends

Write potential outcomes as $C_{g,t}(0)$ without the merger-induced competition change and $C_{g,t}(1)$ with it. Assume (i) the merger affects only the treated group and only in the post period, and (ii) **parallel trends**:

$$\mathbb{E}[C_{T,2}(0) - C_{T,1}(0)] = \mathbb{E}[C_{C,2}(0) - C_{C,1}(0)]. \tag{8}$$

Then

Proposition 1 (DID identifies the average treatment effect on treated)**.** *Under parallel trends,*

$$\mathbb{E}[DID] = \mathbb{E}[C_{T,2}(1) - C_{T,2}(0)].$$

Proof. Expand:

$$\mathbb{E}[DID] = \mathbb{E}[C_{T,2} - C_{T,1}] - \mathbb{E}[C_{C,2} - C_{C,1}] = \mathbb{E}[C_{T,2}(1) - C_{T,1}(0)] - \mathbb{E}[C_{C,2}(0) - C_{C,1}(0)].$$

Add and subtract $\mathbb{E}[C_{T,2}(0)]$:

$$\mathbb{E}[DID] = \mathbb{E}[C_{T,2}(1) - C_{T,2}(0)] + \Big(\mathbb{E}[C_{T,2}(0) - C_{T,1}(0)] - \mathbb{E}[C_{C,2}(0) - C_{C,1}(0)]\Big).$$

The bracketed term is zero by parallel trends, yielding the claim. $\square$

4.3 Model 3: Benchmark-adjusted DID (BDID) via matching

A concern is that treatment and control groups differ in observables (size, value, momentum, baseline coverage). The paper therefore constructs benchmark portfolios (matching cells) using:

$$(\text{size tercile}) \times (\text{BM tercile}) \times (\text{past-return tercile}) \times (\text{coverage tercile}),$$

yielding $3^4 = 81$ cells.

For each treatment stock i, let $BC^i_{C,1}$ and $BC^i_{C,2}$ denote the matched benchmark-cell averages in pre and post. Then the benchmark-adjusted DID for stock i is:

$$BDID_i \equiv (C^i_{T,2} - C^i_{T,1}) - (BC^i_{C,2} - BC^i_{C,1}), \tag{9}$$

and the sample effect is the cross-sectional average of $BDID_i$ across treatment stocks.

Interpretation. $BDID_i$ compares a treatment stock to a *synthetic control* that tracks stocks with similar pre-period observables, reducing the risk that DID captures compositional differences rather than the merger-induced competition change.

4.4 Model 4: Regression DID (panel regression with interaction)

As a robustness/alternative implementation, the paper estimates:

$$C_i = \alpha + \beta_1 MERGE_i + \beta_2 AFFECTED_i + \beta_3(MERGE_i \times AFFECTED_i) + \boldsymbol{\beta}_4' \boldsymbol{Controls}_i + \varepsilon_i, \tag{10}$$

where:

- $MERGE_i$ is an indicator for the post-merger window,
- $AFFECTED_i$ indicates treatment (covered by both houses pre-merger),
- β_3 is the coefficient of interest (the DID effect),
- rich fixed effects are included in practice (merger FE, industry FE, brokerage FE, firm FE).

4.4.1 Why β_3 corresponds to DID

In the simplest 2×2 setting with no controls and group/time means, the difference between post and pre in the treated minus the same difference in controls equals the interaction coefficient. More generally, with controls and fixed effects, β_3 is a conditional DID effect.

5 Mechanism and incentives: career outcomes

5.1 Idea

If competition disciplines bias, then bias should be *less rewarded* when many analysts cover a stock (high competition/peer monitoring). The paper adapts Hong–Kubik style career-outcome regressions and adds an interaction with coverage.

TABLE I
Summary Statistics on the IBES Sample

Variable	Cross-sectional mean (1)	Cross-sectional median (2)	Cross-sectional st. dev. (3)
$\text{COVERAGE}_{i,t}$	21.45	21	9.57
Mean $\text{BIAS}_{i,t}$ (%)	2.70	2.10	3.10
Median $\text{BIAS}_{i,t}$ (%)	2.64	2.01	3.17
Mean $\text{FERROR}_{i,t}$ (%)	3.31	2.39	2.93
Median $\text{FERROR}_{i,t}$ (%)	3.24	2.26	3.00
$\text{FDISP}_{i,t}$(%)	0.75	0.41	1.02
$\text{LNSIZE}_{i,t}$	8.38	8.38	1.62
$\text{SIGMA}_{i,t}$ (%)	40.72	35.04	21.03
$\text{RETANN}_{i,t}$ (%)	1.73	1.49	4.04
$\text{LNBM}_{i,t}$	−1.02	−0.92	0.88
$\text{VOLROE}_{i,t}$ (%)	26.53	10.43	19.79
$\text{PROFIT}_{i,t}$ (%)	15.48	15.29	9.38

5.2 Model 5: linear probability models for promotion/demotion

Using analysts as the unit of observation:

$$Promotion_{i,t+1} = \alpha + \beta_1 Bias_{i,t} + \beta_2 Coverage_{i,t} + \beta_3(Bias_{i,t} \times Coverage_{i,t}) + \boldsymbol{\beta_4' Controls}_{i,t} + \varepsilon_{i,t+1}, \tag{11}$$

$$Demotion_{i,t+1} = \alpha + \beta_1 Bias_{i,t} + \beta_2 Coverage_{i,t} + \beta_3(Bias_{i,t} \times Coverage_{i,t}) + \boldsymbol{\beta_4' Controls}_{i,t} + \varepsilon_{i,t+1}. \tag{12}$$

- $Promotion_{i,t+1} = 1$ if the analyst moves to a larger brokerage house (more analysts employed), 0 otherwise.
- $Demotion_{i,t+1} = 1$ if the analyst moves to a smaller brokerage house, 0 otherwise.
- Controls include forecast accuracy indicators, analyst experience, brokerage size, and fixed effects (year, brokerage, analyst).

Key comparative static. The marginal effect of bias on career outcomes depends on coverage:

$$\frac{\partial Promotion}{\partial Bias} = \beta_1 + \beta_3\, Coverage, \qquad \frac{\partial Demotion}{\partial Bias} = \beta_1 + \beta_3\, Coverage.$$

If competition disciplines bias, one expects $\beta_3 < 0$ in promotion regressions (bias is less rewarded when coverage is higher), and potentially $\beta_3 > 0$ in demotion regressions (bias is more penalized when coverage is higher).

6 Interpreting all tables and figures (table-by-table, figure-by-figure)

6.1 Table I: Summary statistics on the IBES sample

What it is. Cross-sectional averages of key variables in the OLS sample (large firms).
Key takeaways.

TABLE II
REGRESSION OF CONSENSUS FORECAST BIAS ON COMPANY CHARACTERISTICS

	Mean BIAS			Median BIAS		
Variables/model	(1)	(2)	(3)	(4)	(5)	(6)
$COVERAGE_{i,t-1}$	−0.0002**	−0.0002**	0.0001	−0.0002***	−0.0002**	0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
$LNSIZE_{i,t-1}$	0.0028***	0.0026***	0.0044***	0.0028***	0.0026***	0.0042***
	(0.0009)	(0.0009)	(0.0014)	(0.0008)	(0.0008)	(0.0014)
$SIGMA_{i,t-1}$	−0.0093	−0.0031	0.0108***	−0.0095	−0.0061	0.0108***
	(0.0062)	(0.0058)	(0.0039)	(0.0061)	(0.0059)	(0.0041)
$RETANN_{i,t-1}$	−0.1000***	−0.0986***	−0.0368***	−0.0986***	−0.0995***	−0.0367***
	(0.0199)	(0.0199)	(0.0133)	(0.0192)	(0.0198)	(0.0135)
$LNBM_{i,t-1}$	0.0121***	0.0115***	0.0053***	0.0118***	0.0116***	0.0049***
	(0.0016)	(0.0015)	(0.0013)	(0.0016)	(0.0015)	(0.0013)
$VOLROE_{i,t-1}$	0.0062***	0.0061***	0.0000	0.0060***	0.0059***	0.0000
	(0.0019)	(0.0019)	(0.0000)	(0.0019)	(0.0019)	(0.0000)
$PROFIT_{i,t-1}$	0.0579***	0.0571***	0.0629***	0.0578***	0.0574***	0.0619***
	(0.0095)	(0.0092)	(0.0115)	(0.0098)	(0.0095)	(0.0115)
$SP500_{i,t-1}$	−0.0110***	−0.0110***	−0.0017	−0.0110***	−0.0110***	−0.0026
	(0.0024)	(0.0024)	(0.0072)	(0.0025)	(0.0024)	(0.0075)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Brokerage fixed effects	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	No	No	Yes	No	No	Yes
Observations	9,313	9,313	9,313	9,313	9,313	9,313

- Average coverage is large (about 20 analysts), reflecting the focus on the top size quartile.
- Mean and median bias are positive: analysts are optimistic on average.
- Forecast errors are substantial in absolute value, and dispersion in forecasts is nontrivial.

How it connects. This table motivates (i) optimism as a baseline fact and (ii) substantial variation in competition and uncertainty, supporting the need for identification.

6.2 Table II: OLS regression of consensus forecast bias on firm characteristics

$$BIAS_{i,t} = \alpha + \beta\, COVERAGE_{i,t-1} + \boldsymbol{\gamma}'\boldsymbol{X}_{i,t-1} + \lambda_t + \lambda_{\mathrm{ind}(i)} + \varepsilon_{i,t}, \tag{13}$$

Model. OLS regression (13) with mean/median bias as dependent variable.
Reading the coverage coefficient.

- In cross-sectional specifications, $COVERAGE_{t-1}$ loads negatively: more analysts (more competition) is associated with slightly lower bias.
- With firm fixed effects, the coefficient becomes statistically weak and can flip sign, reflecting limited within-firm variation and endogeneity.

Interpretation. The OLS estimates are small in magnitude and unstable across specifications, consistent with coverage being endogenous and **motivating the merger-based design.**

6.3 Table III: Descriptive statistics for mergers

What it is. A catalog of 15 brokerage-house mergers used as competition shocks.
Panel A (overlap stocks). Shows how many stocks each bidder and target covered, and their overlap (the treatment pool).
Panel B (analyst career outcomes). Documents that mergers involve analyst redundancies and

TABLE IV
SUMMARY STATISTICS FOR THE TREATMENT SAMPLE

Variable	Cross-sectional mean (1)	Cross-sectional median (2)	Cross-sectional st. dev. (3)
$\text{COVERAGE}_{i,t}$	21.12	20	9.45
Mean $\text{BIAS}_{i,t}$ (%)	2.79	2.24	3.10
Median $\text{BIAS}_{i,t}$ (%)	2.74	2.21	3.19
Mean $\text{FERROR}_{i,t}$ (%)	3.40	2.52	2.90
Median $\text{FERROR}_{i,t}$ (%)	3.33	2.43	2.99
$\text{FDISP}_{i,t}$ (%)	0.75	0.40	0.94
$\text{LNSIZE}_{i,t}$	8.39	8.37	1.60
$\text{SIGMA}_{i,t}$ (%)	41.00	35.86	21.02
$\text{RETANN}_{i,t}$ (%)	1.74	1.52	4.13
$\text{LNBM}_{i,t}$	−1.03	−0.92	0.91
$\text{VOLROE}_{i,t}$ (%)	25.32	9.89	43.40
$\text{PROFIT}_{i,t}$ (%)	15.52	15.25	9.22

TABLE V
CHANGE IN STOCK-LEVEL COVERAGE AND BIAS: DID ESTIMATOR

Panel A: Coverage and bias	Coverage (1)	Mean BIAS (2)	Median BIAS (3)
SIZE/BM/RET/NOAN-matched	−1.021***	0.0013*	0.0016**
	(0.179)	(0.0007)	(0.0008)

Panel B: Change in forecast bias: DID estimator without analysts from merging houses	Mean BIAS (1)	Median BIAS (2)
SIZE/BM/RET/NOAN-matched	0.0011**	0.0012**
	(0.0005)	(0.0005)

Panel C: Change in forecast bias: Conditioning on initial coverage	Mean BIAS (1)	Median BIAS (2)
SIZE/BM/RET/NOAN-matched (coverage ≤ 5)	0.0078**	0.0096**
	(0.0036)	(0.0044)
SIZE/BM/RET/NOAN-matched (coverage > 5 and ≤ 20)	0.0017*	0.0020*
	(0.0011)	(0.0011)
SIZE/BM/RET/NOAN-matched (coverage > 20)	0.0003	0.0007
	(0.0013)	(0.0013)

exits, consistent with coverage contractions.

Panel C (post-merger coverage sources). Shows which side's analysts continue covering overlapping stocks after the merger.

Why it matters. This table supports the *first-stage premise*: mergers plausibly reduce competition by removing redundant coverage.

6.4 Table IV: Summary statistics for the treatment sample

What it is. Summary stats for stocks in the treatment group (overlap stocks).

Interpretation. Treatment stocks look similar to large-cap OLS stocks in terms of coverage and bias, which is expected because overlap tends to occur for prominent firms.

6.5 Table V: Change in stock-level coverage and bias (DID estimator)

Model. DID/BDID estimators (7)–(9) with matched benchmark portfolios.

Panel A (first stage + reduced form).

- Coverage drops by about one analyst for treatment stocks relative to matched controls.

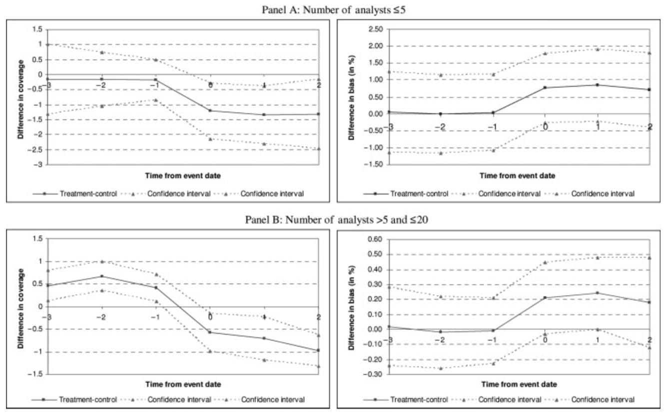

FIGURE I
Trend of Analyst Coverage and Bias in the Treatment Sample (Net of Control)

- Mean and median bias increase in the post window.

Implied causal magnitude (per analyst lost). A simple scaling is

$$\frac{\Delta BIAS}{|\Delta COVERAGE|} \approx \frac{0.0013}{1.021} \approx 0.00127,$$

i.e., roughly **13 basis points of price** more optimism per one-analyst reduction (mean bias).

Panel B (exclude analysts from merging houses). Bias still increases when considering analysts not employed by the merging houses, ruling out a pure *composition* story (e.g., the more optimistic analyst "wins" the job).

Panel C (heterogeneity by initial coverage). The bias increase is strongest for initially low-coverage stocks, consistent with diminishing marginal discipline from extra competitors.

6.6 Figure I: Trend of coverage and bias around merger (net of control)

What it shows. Event-time plots (up to 3 years pre/post) of treatment-minus-control averages.
What to look for.

- No strong pre-trends in bias prior to the merger (supports parallel trends).

- Coverage falls and bias rises after the merger, especially for low initial coverage groups.

6.7 Table VI: Validity of the natural experiment

Panel A (actual earnings). No differential change in realized earnings for treatment vs controls, consistent with the merger affecting *forecasts* rather than fundamentals.
Panel B (firm characteristics). No differential changes in Tobin's Q, size, returns, volatility, profitability, sales, which supports the matching strategy and the exogeneity of the shock with respect to other observables.
Panel C (non-overlapping stocks). Using a "pseudo-treatment" of stocks covered by one but not both merging houses yields no effect, suggesting overlap status is crucial for the mechanism.
Panel D (alternative control group). Using stocks covered by only one merging house as controls still yields positive bias effects, strengthening the baseline result.

TABLE VI
VALIDITY OF THE NATURAL EXPERIMENT

Panel A: Change in earnings

	Mean EARN (1)	Median EARN (2)
DID estimator (SIZE/BM/RET/NOAN-matched)	−0.0002	−0.0002
	(0.0005)	(0.0005)

Panel B: Change in firm characteristics

Stock characteristic	SIZE/BM/RET/NOAN-matched
Tobin's Q	0.0397
	(0.1138)
Size	30.52
	(20.63)
Returns	0.0015
	(0.0012)
Volatility	−0.0069
	(0.0054)
Profitability	−0.0593
	(0.0597)
Log(sales)	0.0046
	(0.0090)

Panel C: Change in forecast bias for non-overlapping stocks

	Coverage (1)	Mean BIAS (2)	Median BIAS (3)
SIZE/BM/RET/NOAN-matched	0.080	0.0001	0.0001
	(0.106)	(0.0004)	(0.0004)

Panel D: Change in forecast bias for non-overlapping stocks as a control

	Mean BIAS (1)	Median BIAS (2)
SIZE/BM/RET/NOAN-matched	0.0017***	0.0018***
	(0.0006)	(0.0006)

6.8 Figure II: Kernel densities of treatment–control bias differences

What it is. Nonparametric densities of (treatment minus control) bias before vs after merger.
Interpretation. The post-merger distribution shifts right, indicating a broad-based increase in optimism bias (not driven by a few outliers). A Kolmogorov–Smirnov test rejects equality of distributions.

6.9 Table VII: Regression evidence for change in forecast bias

Model. Regression DID (10) with fixed effects.
Key coefficient. The interaction $MERGE \times AFFECTED$ is positive and statistically significant, replicating the DID result within a saturated regression framework that controls for:

- merger fixed effects (event-level shocks),
- industry fixed effects,
- brokerage fixed effects,
- firm fixed effects,
- and forecast recency in some columns.

Interpretation. The bias increase is not an artifact of simple mean comparisons; it survives stringent fixed-effect controls.

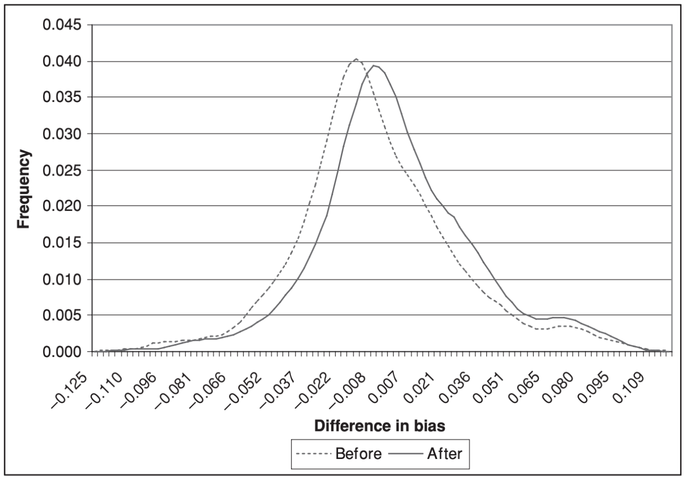

FIGURE II
Kernel Densities of Differences between Treatment and Control before and after Merger

TABLE VII
CHANGE IN FORECAST BIAS: REGRESSION EVIDENCE

	Mean BIAS (1)	Median BIAS (2)	Mean BIAS (3)	Median BIAS (4)
$MERGE_i$	0.0005	0.0005	0.0005	0.0006
	(0.0008)	(0.0008)	(0.0008)	(0.0008)
$AFFECTED_i$	−0.0019**	−0.0019**	−0.0019***	−0.0019***
	(0.0006)	(0.0007)	(0.0006)	(0.0006)
$MERGE_i \times AFFECTED_i$	0.0021*	0.0024**	0.0021*	0.0024**
	(0.0012)	(0.0012)	(0.0012)	(0.0012)
$LNSIZE_{i,t-1}$	0.0038***	0.0037***	0.0038***	0.0037***
	(0.0010)	(0.0010)	(0.0009)	(0.0010)
$RETANN_{i,t-1}$	0.0005	0.0004	0.0005	0.0004
	(0.0017)	(0.0017)	(0.0017)	(0.0017)
$LNBM_{i,t-1}$	−0.0037	0.0001	−0.0037	0.0001
	(0.0073)	(0.0074)	(0.0073)	(0.0074)
$COVERAGE_{i,t-1}$	0.0001	0.0001	0.0001	0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
$SIGMA_{i,t-1}$	0.0001**	0.0000	0.0001**	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
$VOLROE_{i,t-1}$	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
$PROFIT_{i,t-1}$	0.0629***	0.0620***	0.0630***	0.0621***
	(0.0052)	(0.0052)	(0.0052)	(0.0052)
$SP500_{i,t-1}$	0.0032	0.0039	0.0032	0.0039
	(0.0031)	(0.0037)	(0.0031)	(0.0037)
$REC_{i,t-1}$			−0.0000	−0.0000
			(0.0000)	(0.0000)
Merger fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Brokerage fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Observations	57,005	57,005	57,005	57,005

TABLE VIII
CHANGE IN ALTERNATIVE FORECAST BIAS MEASURES: DID ESTIMATOR

	Mean BIAS (1)	Median BIAS (2)
Panel A: Long-term growth		
DID estimator (SIZE/BM/RET/NOAN-matched)	0.553***	0.352*
	(0.202)	(0.212)
Panel B: Analyst recommendations		
DID estimator (SIZE/BM/RET/NOAN-matched)	0.0501	0.0902*
	(0.0412)	(0.0556)

TABLE IX
INCENTIVES AND CAREER OUTCOMES

	Promotion		Demotion	
	(1)	(2)	(3)	(4)
BIAS	0.0106*	0.0448***	−0.0036	−0.0065
	(0.0067)	(0.0136)	(0.0074)	(0.0092)
HIGH ACCURACY	0.0082**	0.0183**		
	(0.0040)	(0.0080)		
LOW ACCURACY			0.0262***	0.0233**
			(0.0062)	(0.0098)
BIAS × COVERAGE		−0.0019***		0.0005*
		(0.0006)		(0.0003)
HIGH ACCURACY × COVERAGE		−0.0006		
		(0.0004)		
LOW ACCURACY × COVERAGE				0.0001
				(0.0005)
COVERAGE	−0.0004*	0.0002	0.0004	0.0002
	(0.0002)	(0.0003)	(0.0003)	(0.0003)
EXPERIENCE	0.0262***	0.0265***	0.0067	0.0065
	(0.0052)	(0.0052)	(0.0043)	(0.0043)
BROKERAGE SIZE	−0.0016***	−0.0016***	0.0008***	0.0008***
	(0.0003)	(0.0003)	(0.0002)	(0.0002)
Year fixed effects	Yes	Yes	Yes	Yes
Brokerage fixed effects	Yes	Yes	Yes	Yes
Analyst fixed effects	Yes	Yes	Yes	Yes
Observations	45,770	45,770	45,770	45,770

6.10 Table VIII: Alternative bias measures (long-term growth; recommendations)

What it tests. Whether reduced competition also increases optimism in other analyst outputs.
Findings.

- Long-term growth forecasts rise after mergers for treatment stocks vs controls.
- Recommendations tilt more positively (higher on the 1–5 scale).

Interpretation. The effect is not limited to one-year EPS forecasts; it appears in broader forward-looking and evaluative analyst signals.

6.11 Table IX: Incentives and career outcomes

Model. Linear probability models (11)–(12) with analyst fixed effects and brokerage/year fixed effects.
Main pattern.

- Promotions: bias is positively associated with promotion, but the interaction $BIAS \times COVERAGE$ is negative, meaning optimism is more rewarded when coverage is low (less competition).

- Demotions: bias is weakly negatively associated with demotion at low coverage, but the interaction is positive, suggesting that in high-coverage environments, optimism can become penalized (standing out as biased when peers disagree).

Mechanism interpretation. This supports an "independence/peer-monitoring" channel: when many analysts cover a stock, deviations from accuracy are more easily detected and less rewarded.

7 Summary

1. **Why OLS is not causal:** coverage is endogenous and persistent.
2. **What the merger design identifies:** a competition shock that reduces coverage for overlap stocks.
3. **What assumptions are needed:** parallel trends + no other merger-correlated shocks differentially affecting treated stocks.
4. **Why matching matters:** overlap stocks are special; BDID reduces observable imbalance.
5. **Where the mechanism evidence comes from:** heterogeneity by initial coverage (Table V, Figure I) and career incentives (Table IX).